

Project Framing

Hospital Chatbot for Color-Coded Clinical Pathways

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Purpose

This pack states what the project is doing: Part 1 (NLP acuity) is built; Part 2 is protocol-mapped colour pathways; Bayesian fusion is out of scope.

1 Project map

One-sentence summary

A hospital chatbot that maps a patient's chief complaint to a SATS acuity level and colour, then attaches a concrete clinical pathway (destination, time target, actions)—without requiring a TEWS+Bayesian fusion stack at intake.

Architecture (implemented)

Chief complaint

- > Medical gate
- > OpenMed NER (optional)
- > BioBERT (5-class acuity)
- > Softmax -> acuity $\hat{c}$ in $\{1..5\}$
- > $C = f_SATS(\hat{c})$ % 1 Red, 2 Orange, 3 Yellow, 4/5 Green
- > Pathway protocol $P(C)$

Part 1 — done

Component	Location
Fine-tuned BioBERT (baseline + SMOTE)	<code>fine-tuned-biobert/</code>
ML API (predict, gate, OpenMed, ...)	<code>curatio/server/ml/</code>
Express gateway	<code>curatio/server/</code>
React demo + presentation UI	<code>curatio/client/</code>
Holdout eval (~99.92% colour accuracy)	<code>results/eval_outputs/</code>
Results slides	<code>results/slides/</code>

Colour mapping in code: `curatio/server/ml/predict.py` (SATS_BY_ACUITY).

Part 2 — implemented

Pathways $P(C)$ attach to **fused** colour from `/fuse`. TEWS, rule discriminators, and tabular Bayes vote under max-urgency f_{fusion} (full BN learning remains out of scope). Spec: `triage-fusion/`.

Explicit non-goals

- No full Bayesian network (pgmpy / structure learning)
- No claim that BioBERT itself performs TEWS–Bayes fusion
- No prospective clinical trial in this draft

2 Research gaps

1. **Acuity discontinuity** — colour does not drive care after the desk; answer with pathway object $P(C)$.
2. **LMIC / Ghanaian chief-complaint NLP** — local text→SATS colour via BioBERT + Curatio.
3. **Green Tag Burden** — Green pathway diversion to OPD / Minors / Polyclinic.
4. **Constrained classifier vs open LLM** — fixed 5-class head + deterministic maps.
5. **Text-only pre-triage** — NLP acuity before vitals; optional TEWS when supplied via `/fuse`.
6. **Minority Red safety** — baseline vs SMOTE; report L1 recall.
7. **Colour must mean a pathway** — not only a badge.

3 Ten research questions

- RQ1** Can unstructured chief complaints map to SATS acuity 1–5 with usable accuracy?
- RQ2** Does correct acuity imply correct colour under fixed f_{SATS} ?
- RQ3** How should colours become pathways (destination, T_{max} , actions)?
- RQ4** Can a chatbot deliver pre-triage acuity and pathway guidance?
- RQ5** How well does the system reject non-clinical input?
- RQ6** Does OpenMed NER improve interpretability without replacing acuity?
- RQ7** How does imbalance affect Level-1 (Red) recall?
- RQ8** How overconfident is softmax, and what about nurse oversight?
- RQ9** Can Green diversion be specified to reduce ED congestion?
- RQ10** Does f_{fusion} assign a safer colour than C_{NLP} when TEWS and language disagree?
- RQ11** Are pathway cards $P(C)$ correctly attached to fused colours in API tests?

Formal Part 1: Input $X \rightarrow \hat{c} \in \{1, \dots, 5\} \rightarrow C = f_{\text{SATS}}(\hat{c})$.

Formal Part 2: $C \mapsto P(C) = (T_{\text{max}}, \text{dest}, \text{actions}, \text{escalation})$.

4 Pathways design (no Bayesian)

Colour	$T_{\max}$	Destination	Actions (summary)
Red	0 min	Resuscitation	Code Red; bypass registration
Orange	10 min	Acute bed	Countdown; escalate if unclaimed
Yellow	60 min	Urgent ED stream	Standing orders in parallel
Green	Routine	OPD / Minors / Polyclinic	Divert from acute ED
Blue	Dignity	Mortuary pathway	Silent social-work protocol

5 Mathematics scope

In scope: softmax acuity; f_{SATS} ; pathway $P(C)$; evaluation metrics; medical gate.

Out of scope: TEWS sum engine; Bayesian $P(C_k | E)$; fusion max-urgency controller.

6 Thesis outline

- Ch.1 Introduction — problem, 10 RQs, scope (no Bayes)
- Ch.2 Literature — SATS, Green Tag, triage NLP
- Ch.3 Methodology — BioBERT, colour map, pathways
- Ch.4 Results — holdout metrics, calibration, scenarios
- Ch.5 Conclusion — RQ answers, future TEWS overlay (optional)

Location: `thesis/first-draft/`.

7 Roadmap

- Done: Part 1 software/eval; framing; findings; thesis first draft; no-Bayes decision
- Write: deepen Ch.2 from reading notes; expand bib
- Optional build: pathway card fields in API/UI
- Do not schedule: Bayesian networks / fusion controller